

MOBILE APP- FARM MANAGEMENT

(AGRO SEVA)

Submitted to

THE GANDHIGRAM RURAL INSTITUTE (Deemed to University)

In Partial fulfillment of the requirement for the award of the Degree of

MASTER OF COMPUTER APPLICATIONS

BY

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DEPARTMENT OF COMPUTER SCIENCE AND APPLICATIONS

THE GANDHIGRAM RURAL INSTITUTE

(DEEMED TO BE UNIVERSITY)

(Ministry of Human Resource Development, Government of India)

Accredited NAAC with 'A' Grade (3rd Cycle)

Gandhigram – 624302

DECEMBER 2024

BONAFIDE CERTIFICATE

This is to certify that the project titled “**MOBILE APPLICATION – FARM MANAGEMENT (AGRO SEVA)**” is a Bonafide record of work carried out by **S. RAVIN RAJ (23322009), E. NIVETHA (23322022)** submitted in partial fulfillment of the requirement for the award of the Degree of **Master of Computer Applications (MCA)** in The Gandhigram Rural Institute (Deemed to be University), Gandhigram, during the July 2024 - December 2024.

Dr.M. Mary Shanthi Rani, MCA, M.Phil., Ph. D

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Internal Guide

Head of the Department

Submitted for the Viva-Voice examination held on_____

Signature of the Examiner(s)

DECLARATION

I hereby declare that this mini project (Mobile Apps for Agriculture) Work titled “**MOBILE APPLICATION – FARM MANAGEMENT (AGRO SEVA)**” is a record of original work done by me Under the supervision & guidance of Dr. **M. Mary Shanthi Rani, MCA, M.Phil., Ph.D.**, Professor, Department of Computer Science and Applications. The project work has not formed the basis for the award of any Degree / Diploma / Associate ship / Fellowship or Similar title to any candidate of any other University.

Place: Gandhigram

S. RAVIN RAJ (23322009)

Date:

E. NIVETHA (23322022)

ACKNOWLEDGEMENT

First and foremost, I would like to thank the Almighty for blessing me with the physical and mental strength for doing this project. I sincerely thank **Dr.M. Mary Shanthi Rani, MCA, Ph.D.**, Professor, Department of Computer Science and Applications, **The Gandhigram Rural Institute (Deemed to be University), Gandhigram** for the motivation and valuable suggestions given to me to do. This project work. She has been a constant source of inspiration to me and her Support has encouraged me in the successful completion of project in time.

I sincerely thank the Head of the Department **Dr.P. Kalavathi, MCA, M.Phil., Ph.D.**, Faculty members of the department, Research Scholars and the non-teaching staff of the department for their help and support to complete this project.

It is also my duty express my heartfelt thanks to my parents, friends and those who have directly and in directly helped me in completing this Project work.

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CHAPTER I

INTRODUCTION

INTRODUCTION

1.1 ABSTRACT

Farm Management (Agro-Seva) is a dedicated mobile application designed to assist farmers in cultivating tomato plants effectively. The app provides step-by-step guidance and features a secure login system to personalize the user experience. Its home page offers easy navigation to all tools and functionalities, ensuring a smooth user experience. Farmers can document their daily farming activities using the day-by-day activity tracker, with all information securely stored in a database for future reference.

A built-in profit-and-loss calculator allows farmers to analyze their financial performance and make informed decisions for better profitability. The app also integrates soil testing, enabling users to assess soil quality and receive tailored recommendations to optimize crop yield. Designed for accessibility, Farm Management (Agro-Seva) supports Tamil and English languages, catering to a diverse farmer base.

With its combination of data management, financial insights, and soil analysis, the app empowers farmers to adopt modern, efficient farming practices. It acts as a comprehensive tool to monitor progress, address challenges, and enhance overall productivity in tomato cultivation.

CHAPTER II

SYSTEM SPECIFICATIONS

HARDWARE SPECIFICATION:

RAM	:	16GB
Processor	:	Intel Core i5 13 th Gen
HardDisk	:	512GB SSD
Monitor	:	15.1” Color
System Type	:	64-bitOperatingSystem
Processor Speed	:	2.60GHz

SOFTWARE SPECIFICATION:

Programming Tool	:	Android Studio
Programming Languages	:	Java, XML
Testing Tools	:	Android Emulator
Operating Tools	:	Windows 11 Home Single

CHAPTER III

SOFTWARE DESCRIPTION

SOFTWARE DESCRIPTION:

ANDROID STUDIO:



Android Studio is an Integrated Development Environment (IDE) designed by Google for developing Android applications. Based on IntelliJ IDEA, it offers robust tools for creating high-quality apps for Android devices. It supports Java, Kotlin, and C++ programming languages, providing developers with flexibility. Features like a rich code editor, intelligent code completion, and refactoring tools streamline the coding process. The built-in emulator, along with support for real-time preview of UI designs, accelerates the development and testing cycle. Android Studio includes features like Gradle-based build automation, APK analysis, and performance profiling tools to optimize application performance. It also provides support for advanced development needs, such as creating apps for Android TV, Wear OS, and Android Auto. With frequent updates and direct integration with Firebase for app analytics, crash reporting, and real-time database management, Android Studio simplifies and enhances the app development experience. It is the go-to tool for developers looking to create innovative and efficient Android applications.

JAVA:



Java is a high-level, object-oriented programming language developed by Sun Microsystems in 1995, now owned by Oracle Corporation. It is widely used due to its platform independence, achieved through the Java Virtual Machine (JVM), which allows programs to run on any device with a compatible JVM. Java's syntax is simple, robust, and similar to C++, making it accessible for beginners and powerful enough for experts. Its features, such as garbage collection, exception handling, and multithreading, contribute to developing secure, efficient, and scalable applications.

Java is used across diverse domains, from building mobile apps with Android, enterprise applications with Java EE, web applications, and even embedded systems. Its extensive libraries, frameworks like Spring and Hibernate, and an active developer community provide tools for solving a wide range of programming challenges. Furthermore, Java's backward compatibility ensures that programs written in older versions run seamlessly on newer ones, securing its position as a reliable choice for software development.

XML:



XML (Extensible Markup Language) is a foundational tool used in Android Studio for designing user interfaces. It serves as the blueprint for defining the layout and structure of Android app screens, enabling developers to create visually appealing and functional interfaces. XML files in Android Studio specify the arrangement of UI components such as buttons, text fields, images, and containers in a hierarchical structure. With attributes like size, position, colour, and style, developers can precisely control the appearance and behaviour of each element. XML layouts are intuitive and easy to modify, providing a declarative way to design the app's user interface without requiring deep knowledge of coding logic for initial setup.

What makes XML particularly effective in Android Studio is its seamless integration with Java/Kotlin code through resource IDs. Developers can dynamically manipulate the XML-defined UI components during runtime, adding interactivity and responsiveness to the app. Additionally, Android Studio offers a visual layout editor that simplifies working with XML by allowing drag-and-drop functionality, providing a real-time preview of the design across different screen sizes and orientations. This combination of XML's simplicity and the powerful tools in Android Studio enables developers to create consistent, adaptive, and polished user interfaces for a wide range of Android devices.

WINDOWS 11:



Windows 11, developed by Microsoft, is a versatile operating system released in July 2015. Designed for both personal and professional use, it combines user-friendly features with robust functionality. Its seamless interface caters to various devices, including desktops, laptops, tablets, and even hybrid devices. The operating system introduced features like the Start Menu's return, Cortana digital assistant, virtual desktops, and a touch-friendly interface, making it adaptable for different use cases. Its modular updates ensure enhanced security and functionality over time.

Windows 11 integrates tools like Microsoft Edge, DirectX 12 for gaming, and enhanced accessibility options to boost productivity and usability. It supports applications across the Microsoft Store and legacy software, ensuring compatibility with a vast range of programs. Security features such as Windows Defender, BitLocker encryption, and regular updates make it a secure choice for users. Additionally, its enterprise features, like remote desktop and domain join, make it a popular operating system for business environments worldwide.

FIREBASE:



Firebase evolved from Envolv. Envolv is a prior startup founded by James Tamplin and Andrew Lee in 2011. Envolv provided developers an API which allowed the integration of online chat functionality into their websites. After releasing the chat service, it found that the envlove was being used to pass application data, which were not chat messages. Developers used Envolv to sync application to separate the real-time architecture and the chat system which powered it. In September 2011, Tamplin and Lee founded firebase as a separate company. It was lastly launched to the public in April 2012.

Firebase Real-time Database was the first product of firebase. It is an API which syncs application data across Android, iOS, and Web devices. It gets stored on Firebase's cloud. Then the firebase real-time database helps the developers to build real-time, collaborative applications.

CHAPTER IV

PROJECT AND MODULE DESCRIPTION

SYSTEM DESIGN

4.1 PROJECT DESCRIPTION

Agriculture is the backbone of the economy in many regions, yet farmers face numerous challenges in managing crop cultivation effectively. **Farm Management (AgroSeva)** addresses these issues by offering a user-friendly mobile application tailored specifically for tomato farming. The app provides a streamlined sign-up and login process, including a forget-password option for ease of use. Upon logging in, users are greeted with a categorized home page that separates options for vegetables and fruits.

The **tomato farming module** is packed with features designed to simplify and enhance agricultural practices. It includes step-by-step cultivation instructions to guide farmers through every stage, soil testing tools to determine optimal growth conditions, and a water management system with reminders for scheduling irrigation by date and time. The app also supports labour management, allowing farmers to allocate resources effectively, and features a profit/loss analysis tool for better financial decision-making.

Additionally, AgroSeva incorporates a **daily rate module** to track market trends and pricing, along with a weather API that provides real-time updates to help farmers plan their activities. With multilingual functionality, the app ensures accessibility for a wide range of users, catering to diverse linguistic backgrounds.

By integrating advanced technology into traditional farming, AgroSeva bridges the gap between conventional agriculture and modern digital solutions. This holistic approach empowers farmers to increase their yields, streamline operations, and improve their overall financial outcomes. AgroSeva stands as a vital tool in driving innovation and sustainability in agriculture, making farming smarter and more efficient.

4.2 MODULE DISCRIPTION

LIST OF MODULES:

- Signup and Login
- Home Page
- Fruits Module
- Offer Notification
- Language Translator
- Weather
- Rate

SIGNUP AND LOGIN

Allows users to create an account and securely log in to the app.

Includes a "Forget Password" option for account recovery.

HOME PAGE

Categorized into Vegetables and Fruits.

Provides a clear navigation structure for users.

FRUITS MODULE

a) Tomato Fruits

i) This section is dedicated to tomato farming, comprising five subcategories:

b) Instructions:

i) Provides a comprehensive step-by-step guide to tomato cultivation, covering soil preparation, planting, irrigation, pest management, and harvesting.

c) Soil Test:

Accepts inputs for soil parameters like pH level, Potassium (K), Phosphorus (P), and Nitrogen (N).

Analyzes and determines if the soil is suitable for tomatoes. o Suggests necessary fertilizers and soil amendments based on the analysis.

d) Water Reminder:

Sends timely notifications to remind users to water their plants, ensuring optimal growth.

e) Labour Tracking:

Enables farmers to store details of labourers, including names, tasks, and salaries. o Helps in maintaining a clear record of workforce management.

f) Profit and Loss:

Facilitates financial analysis by calculating profit or loss after inputting all farming-related data, such as expenses and yield.

LANGUAGE TRANSLATOR

Provides translation support for Tamil and English, ensuring the app is accessible to users from different linguistic backgrounds.

OFFER NOTIFICATION

The Offer Notification feature keeps farmers updated on the latest agricultural deals, discounts, and government schemes. It ensures timely alerts to help farmers save costs and access valuable resources easily.

WEATHER

The app integrates a **weather API** to provide real-time updates on climatic conditions, helping farmers plan their activities effectively. With accurate forecasts, users can optimize irrigation schedules, protect crops from adverse weather, and ensure better yield management. This feature empowers farmers to make informed decisions and minimize risks associated with unpredictable weather.

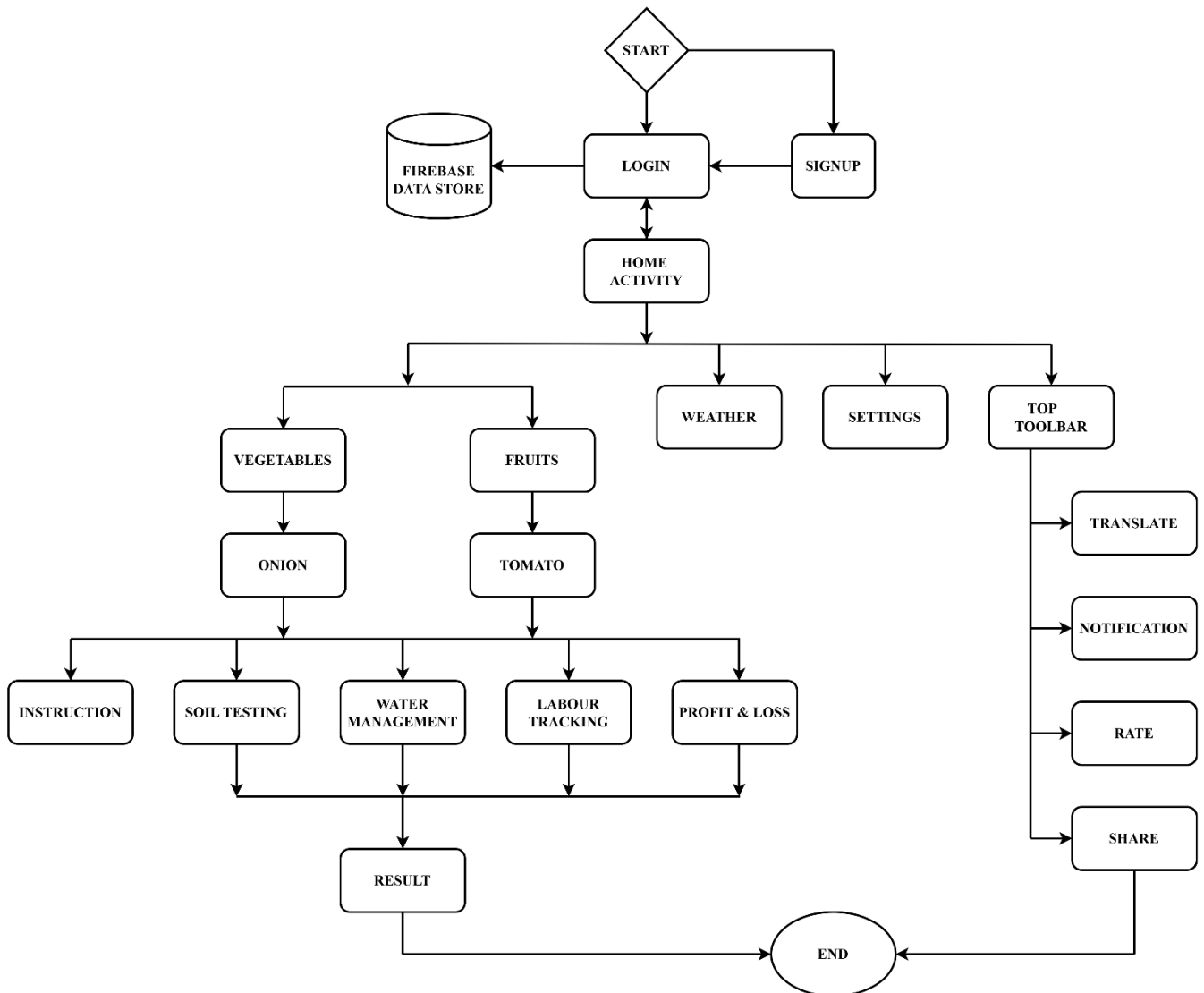
RATE

The app features a **daily rate module** that provides real-time updates on market prices, helping farmers stay informed about current trends. This allows users to make strategic decisions about selling their produce at the best possible rates. By tracking market fluctuations, farmers can maximize their profits and plan their sales effectively.

CHAPTER V

DATA FLOW DIAGRAM

DATA FLOW DIAGRAM



CHAPTER VI

TESTING

TESTING:

FUNCTIONAL TESTING:

Ensures that all features and modules, such as the Crop Selection Module, Signup and Login Module, Home Page Module, Weather API, Rate API, Language Translator Module, and Profit and Loss Calculation Module for labor management, work as intended.

USABILITY TESTING:

Verifies that the app provides a user-friendly interface and smooth navigation. It evaluates the ease of accessing weather updates, market rate information, and labor management tools to ensure farmers can manage their operations efficiently.

PERFORMANCE TESTING:

Measures the app's performance under various conditions, such as high user loads during peak farming seasons. It ensures stability, quick loading, and the ability to handle simultaneous access to weather data, market rates, and financial calculations.

COMPATIBILITY TESTING:

Tests the app across multiple devices, screen sizes, and operating systems to ensure a consistent experience, including seamless functionality of APIs and multilingual support.

SECURITY TESTING:

Checks for vulnerabilities in user data handling, financial calculations, and the integration of APIs to ensure secure transactions, data protection, and the confidentiality of farming-related records.

REGRESSION TESTING:

Performed after updates or bug fixes to confirm that key features like weather updates, rate tracking, language options, and profit/loss calculations continue to function correctly without introducing new issues.

USER ACCEPTANCE TESTING:

Involves farmers or stakeholders testing the app in real-world scenarios to validate that the weather and rate APIs, multilingual support, and labor management tools meet the practical needs of farming operations.

BETA TESTING:

Conducted by releasing the app to a select group of farmers to identify any real-world issues in functionalities such as market rate updates, weather forecasting, and profit/loss analysis before the full launch.

CHAPTER VII

APPENDIX

APPENDIX:

APPENDIX A – SAMPLE SOURCECODE

AndroidManifest.xml:

```
<?xml version="1.0" encoding="utf-8"?>
  <manifest xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:tools="http://schemas.android.com/tools">
    <uses-permission android:name="android.permission.INTERNET"/>
    <application android:allowBackup="true"
      android:dataExtractionRules="@xml/data_extraction
_rules"
      android:fullBackupContent="@xml/backup_rules"
      android:icon="@mipmap/ic_launcher"
      android:label="@string/app_name"
      android:roundIcon="@mipmap/ic_launcher_round"
      android:supportsRtl="true"
      android:theme="@style/Base.Theme.AgroSeva"
      tools:targetApi="31">
      <activity
        android:name=".PassswordActivity"
        android:exported="false" />
      <activity
        android:name=".HomeActivity"
        android:exported="false" />
      <activity
        android:name=".SignupActivity"
        android:exported="false" />
      <activity
        android:name=".LoginActivity"
        android:exported="false" />
      <activity
        android:name=".SplashActivity"
```

```

        android:exported="true">
        <intent-filter>

        <action android:name="android.intent.action.MAIN" />

        <category android:name="android.intent.category.LAUNCHER" />
        </intent-filter>
    </activity>
</application>

</manifest>

```

Activity_home.xml:

```

<?xml version="1.0" encoding="utf-8"?>
<androidx.coordinatorlayout.widget.CoordinatorLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:background="@drawable/background_image">

    <com.google.android.material.appbar.AppBarLayout
        android:layout_width="match_parent"
        android:layout_height="wrap_content">

        <androidx.appcompat.widget.Toolbar
            android:id="@+id/toolbar"
            android:layout_width="match_parent"
            android:layout_height="?attr/actionBarSize"
            app:title="@string/app_name"
            app:titleTextColor="@color/color_white"
            android:background="@color/green"
            app:popupTheme="@style/ThemeOverlay.AppCompat.Light"
            app:menu="@menu/top_menu" />

```

```
</com.google.android.material.appbar.AppBarLayout>
```

```
<FrameLayout
```

```
    android:id="@+id/framelayout"
```

```
    android:layout_width="match_parent"
```

```
    android:layout_height="match_parent"
```

```
    android:layout_marginBottom="80dp"
```

```
    app:layout_behavior="@string/appbar_scrolling_view_behavior">
```

```
</FrameLayout>
```

```
<com.google.android.material.bottomnavigation.BottomNavigationView
```

```
    android:id="@+id/bottomnavbar"
```

```
    android:layout_width="match_parent"
```

```
    android:layout_height="wrap_content"
```

```
    android:layout_gravity="bottom"
```

```
    app:menu="@menu/bottom_navigation_menu"
```

```
    android:background="@color/green" />
```

```
</androidx.coordinatorlayout.widget.CoordinatorLayout>
```

HomeActivity.java:

```
package com.example.agroseva;
```

```
import android.annotation.SuppressLint;
```

```
import android.app.NotificationChannel;
```

```
import android.app.NotificationManager;
```

```
import android.content.Intent;
```

```
import android.content.pm.PackageManager;
```

```
import android.net.Uri;
```

```
import android.os.Build;
```

```
import android.os.Bundle;
```

```
import android.provider.Settings;
```

```
import android.view.Menu;
```

```
import android.view.MenuItem;
```

```

import android.webkit.WebViewClient;
import android.widget.Toast;
import android.Manifest;

import androidx.annotation.NonNull;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.content.ContextCompat;
import androidx.fragment.app.Fragment;
import androidx.work.Configuration;

import com.example.agroseva.databinding.ActivityHomeBinding;

import java.util.Locale;

public class HomeActivity extends AppCompatActivity {

    private static final int PERMISSION_REQUEST_CODE = 1001;
    ActivityHomeBinding binding;

    @SuppressWarnings("NonConstantResourceId")
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        getWindow().setStatusBarColor(ContextCompat.getColor(HomeActivity.this, R.color.green));

        binding = ActivityHomeBinding.inflate(getLayoutInflater());
        setContentView(binding.getRoot());

        // Create Notification Channel (required for Android Oreo and above)
        if (Build.VERSION.SDK_INT >= Build.VERSION_CODES.O) {
            createNotificationChannel();
        }

        // Set up the toolbar

```

```

setSupportActionBar(binding.toolbar);

// Load the default fragment
replaceFragment(new HomeFragment());

// Set up the bottom navigation view
binding.bottomnavbar.setBackground(null);
binding.bottomnavbar.setOnItemSelectedListener(item -> {
    int itemId = item.getItemId();
    if (itemId == R.id.home) {
        replaceFragment(new HomeFragment());
        Toast.makeText(this, "Home", Toast.LENGTH_SHORT).show();
    } else if (itemId == R.id.weather) {
        replaceFragment(new WeatherFragment());
        Toast.makeText(this, R.string.weather, Toast.LENGTH_SHORT).show();
    } else if (itemId == R.id.settings) {
        replaceFragment(new SettingsFragment());
        Toast.makeText(this, "Settings", Toast.LENGTH_SHORT).show();
    }
    return true;
});
}

private void replaceFragment(Fragment fragment) {
    getSupportFragmentManager()
        .beginTransaction()
        .replace(R.id.framelayout, fragment) // Ensure this ID matches your layout
        .commit();
}

private void createNotificationChannel() {
    CharSequence name = "Watering Reminder Channel";
    String description = "Channel for watering reminders";
    int importance = NotificationManager.IMPORTANCE_HIGH;

```

```

NotificationChannel channel = null;
if (Build.VERSION.SDK_INT >= Build.VERSION_CODES.O) {
    channel = new NotificationChannel("watering_channel", name, importance);
}
if (Build.VERSION.SDK_INT >= Build.VERSION_CODES.O) {
    channel.setDescription(description);
}

```

```

NotificationManager notificationManager = getSystemService(NotificationManager.class);
if (notificationManager != null) {
    if (Build.VERSION.SDK_INT >= Build.VERSION_CODES.O) {
        notificationManager.createNotificationChannel(channel);
    }
}
}

```

```

@Override
public boolean onCreateOptionsMenu(Menu menu) {
    getMenuInflater().inflate(R.menu.top_menu, menu);
    return true;
}

```

```

@Override
public boolean onOptionsItemSelected(@NonNull MenuItem item) {
    int itemId = item.getItemId();
    if (itemId == R.id.rate) {
        Intent intent = new Intent(Intent.ACTION_VIEW,
Uri.parse("https://vegetablemarketprice.com/market/tamilnadu/today"));
        startActivity(intent);
    } else if (itemId == R.id.notification) {
        replaceFragment(new NotificationFragment());
        Toast.makeText(this, "Notification", Toast.LENGTH_SHORT).show();
        return true;
    } else if (itemId == R.id.share) {

```



```

Intent shareIntent = new Intent(Intent.ACTION_SEND);
shareIntent.setType("text/plain");
String shareMessage = "Check out this amazing app:
https://play.google.com/store/apps/details?id=com.example.agroseva";
shareIntent.putExtra(Intent.EXTRA_TEXT, shareMessage);
startActivity(Intent.createChooser(shareIntent, "Share via"));
Toast.makeText(this, "Share", Toast.LENGTH_SHORT).show();
return true;
} else if (itemId == R.id.action_translate) {
    toggleLanguage();
    return true;
}
return super.onOptionsItemSelected(item);
}

private void toggleLanguage() {
    String currentLanguage = getResources().getConfiguration().locale.getLanguage();
    String newLanguage = currentLanguage.equals("en") ? "ta" : "en";

    // Update app's locale
    Locale locale = new Locale(newLanguage);
    Locale.setDefault(locale);
    android.content.res.Configuration config = new android.content.res.Configuration();
    config.setLocale(locale);

    getBaseContext().getResources().updateConfiguration(config,
getBaseContext().getResources().getDisplayMetrics());

    // Notify user
    String languageChangedMessage = newLanguage.equals("en") ? "Language changed to English" :
"மொழி தமிழாக மாற்றப்பட்டது";
    Toast.makeText(this, languageChangedMessage, Toast.LENGTH_SHORT).show();

    // Refresh UI

```

```

        recreate();
    }

    @Override
    public void onRequestPermissionsResult(int requestCode, @NonNull String[] permissions, @NonNull
int[] grantResults) {
        super.onRequestPermissionsResult(requestCode, permissions, grantResults);

        if (requestCode == PERMISSION_REQUEST_CODE) {
            if (grantResults.length > 0 && grantResults[0] == PackageManager.PERMISSION_GRANTED)
            {
                Toast.makeText(this, "Notification permission granted", Toast.LENGTH_SHORT).show();
            } else {
                Toast.makeText(this, "Notification permission denied", Toast.LENGTH_SHORT).show();
            }
        }
    }
}

```

fragment_home.xml:

```

<?xml version="1.0" encoding="utf-8"?>
<ScrollView xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:background="@drawable/background_image"
    tools:context=".HomeFragment">

    <!-- GridLayout for displaying items -->
    <LinearLayout
        android:layout_width="match_parent"
        android:layout_height="match_parent"
        android:padding="8dp"

```

```
android:orientation="vertical"
android:gravity="center"
android:layout_marginTop="50dp">
```

```
<TextView
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="@string/products"
    android:textSize="18sp"
    android:textColor="@color/green"
    android:gravity="center"
    android:textStyle="bold"
    android:layout_gravity="center" />
```

```
<androidx.cardview.widget.CardView
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_margin="20dp"
    app:cardCornerRadius="16dp"
    app:cardElevation="4dp">
```

```
<LinearLayout
    android:id="@+id/vegetables_linear"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:orientation="vertical"
    android:padding="8dp">
```

```
<ImageView
    android:layout_width="300dp"
    android:layout_height="120dp"
    android:src="@drawable/vegetables"
    android:contentDescription="@string/vegetables"
    android:layout_marginBottom="8dp"
```

```
    android:scaleType="fitCenter"
    android:layout_gravity="center" />
```

```
<TextView
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="@string/vegetables"
    android:textSize="18sp"
    android:textColor="@color/black"
    android:gravity="center"
    android:layout_gravity="center" />
```

```
</LinearLayout>
```

```
</androidx.cardview.widget.CardView>
```

```
<androidx.cardview.widget.CardView
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_margin="20dp"
    app:cardCornerRadius="16dp"
    app:cardElevation="4dp">
```

```
<LinearLayout
    android:id="@+id/fruits_linear"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:orientation="vertical"
    android:padding="8dp">
```

```
<ImageView
    android:layout_width="300dp"
    android:layout_height="120dp"
    android:src="@drawable/fruits"
    android:contentDescription="@string/fruits"
    android:layout_marginBottom="8dp"
```

```
        android:scaleType="fitCenter"
        android:layout_gravity="center" />
```

```
    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="@string/fruits"
        android:textSize="18sp"
        android:textColor="@color/black"
        android:gravity="center"
        android:layout_gravity="center" />
```

```
    </LinearLayout>
```

```
</androidx.cardview.widget.CardView>
```

```
</LinearLayout>
```

```
</ScrollView>
```

HomeFragment.java:

```
package com.example.agroseva;
```

```
import android.os.Bundle;
import android.view.LayoutInflater;
import android.view.View;
import android.view.ViewGroup;
import android.widget.LinearLayout;
import android.widget.TextView;
```

```
import androidx.fragment.app.Fragment;
import androidx.fragment.app.FragmentManager;
import androidx.fragment.app.FragmentTransaction;
```

```
public class HomeFragment extends Fragment {
```

```
    @Override
```

```

public View onCreateView(LayoutInflater inflater, ViewGroup container,
                        Bundle savedInstanceState) {
    // Inflate the layout for this fragment
    View view = inflater.inflate(R.layout.fragment_home, container, false);

    // Initialize all LinearLayouts and set click listeners
    setupFragmentNavigation(view.findViewById(R.id.vegetables_linear));
    setupFragmentNavigation(view.findViewById(R.id.fruits_linear));

    return view;
}

private void setupFragmentNavigation(LinearLayout linearLayout) {
    linearLayout.setOnClickListener(v -> {
        // Get the TextView inside the LinearLayout
        TextView textView = (TextView) linearLayout.getChildAt(1); // Assuming TextView is the
second child
        String fragmentName = textView.getText().toString();

        // Open the corresponding fragment
        openFragment(fragmentName);
    });
}

private void openFragment(String fragmentName) {
    Fragment fragment;

    // Match fragment name to corresponding Fragment class
    if (fragmentName.equals("Vegetables")) {
        fragment = new VegetablesFragment();
    } else if (fragmentName.equals("Fruits")) {
        fragment = new FruitsFragment();
    } else {

```

```

        throw new IllegalArgumentException("Unknown fragment name: " + fragmentName);
    }

    // Perform the fragment transaction
    FragmentManager fragmentManager = requireActivity().getSupportFragmentManager();
    FragmentTransaction transaction = fragmentManager.beginTransaction();
    transaction.replace(R.id.frame_layout, fragment); // Replace with your container ID
    transaction.addToBackStack(null); // Optional: add to backstack
    transaction.commit();
}
}

```

fragment_weather.xml:

```

<?xml version="1.0" encoding="utf-8"?>
<ScrollView xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:background="@drawable/background_image"
    tools:context=".HomeFragment">

    <LinearLayout
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:orientation="vertical"
        android:padding="16dp">

        <!-- Existing sections -->

        <!-- Weather Condition Section -->
        <androidx.cardview.widget.CardView
            android:layout_width="match_parent"
            android:layout_height="wrap_content"

```

```
android:layout_margin="8dp"
app:cardCornerRadius="16dp"
app:cardElevation="4dp">
```

```
<LinearLayout
```

```
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:orientation="vertical"
    android:padding="8dp">
```

```
<!-- Spinner for selecting cities -->
```

```
<TextView
```

```
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="@string/weather_select_city"
    android:textSize="18sp"
    android:layout_gravity="center"
    android:textColor="@color/green"
    android:textStyle="bold"
    android:layout_marginBottom="8dp" />
```

```
<Spinner
```

```
    android:id="@+id/citySpinner"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:entries="@array/tamil_nadu"
    android:padding="12dp"
    android:layout_marginBottom="12dp" />
```

```
<!-- Display Weather Info -->
```

```
<TextView
```

```
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="@string/weather_info_label"
```



```
    android:textColor="@color/green"
    android:textSize="18sp"
    android:layout_gravity="center"
    android:textStyle="bold"
    android:layout_marginBottom="8dp" />
```

```
<TextView
    android:id="@+id/weatherInfo"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="@string/weather_default_info"
    android:textSize="16sp"
    android:layout_marginBottom="16dp" />
```

```
<!-- Button to fetch weather -->
```

```
<Button
    android:id="@+id/getWeatherButton"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="@string/get_weather_button"
    android:layout_gravity="center"
    android:backgroundTint="@color/green"
    android:textColor="@color/color_white" />
```

```
</LinearLayout>
```

```
</androidx.cardview.widget.CardView>
```

```
</LinearLayout>
```

```
</ScrollView>
```

WeatherFragment.java:

```
package com.example.agroseva;
```

```
import android.os.Bundle;
import android.view.LayoutInflater;
import android.view.View;
import android.view.ViewGroup;
import android.widget.Button;
import android.widget.Spinner;
import android.widget.TextView;
import android.widget.Toast;
```

```
import androidx.fragment.app.Fragment;
```

```
import org.json.JSONObject;
```

```
import java.io.BufferedReader;
import java.io.InputStreamReader;
import java.net.HttpURLConnection;
import java.net.URL;
```

```
public class WeatherFragment extends Fragment {
```

```
    private static final String WEATHER_API_KEY = "bb513cafd6a336940e677f7f200a4009"; // Replace
    with your OpenWeatherMap API Key
```

```
    public WeatherFragment() {
        // Required empty public constructor
    }
```

```
@Override
```

```
public View onCreateView(LayoutInflater inflater, ViewGroup container, Bundle savedInstanceState) {
    return inflater.inflate(R.layout.fragment_weather, container, false);
```

```
}
```

```
@Override
```

```
public void onCreateView(View view, Bundle savedInstanceState) {
```

```
    Spinner citySpinner = view.findViewById(R.id.citySpinner);
```

```
    Button getWeatherButton = view.findViewById(R.id.getWeatherButton);
```

```
    TextView weatherInfo = view.findViewById(R.id.weatherInfo);
```

```
    getWeatherButton.setOnClickListener(v -> {
```

```
        String selectedCity = citySpinner.getSelectedItem().toString();
```

```
        fetchWeatherData(selectedCity, weatherInfo);
```

```
    });
```

```
}
```

```
private void fetchWeatherData(String city, TextView weatherInfo) {
```

```
    new Thread(() -> {
```

```
        try {
```

```
            // Create the API URL
```

```
            String apiUrl = "https://api.openweathermap.org/data/2.5/weather?q=" + city + "&appid=" +  
WEATHER_API_KEY + "&units=metric";
```

```
            // Open the connection
```

```
            URL url = new URL(apiUrl);
```

```
            HttpURLConnection connection = (HttpURLConnection) url.openConnection();
```

```
            connection.setRequestMethod("GET");
```

```
            // Read the response
```

```
            BufferedReader reader = new BufferedReader(new  
InputStreamReader(connection.getInputStream()));
```

```
            StringBuilder response = new StringBuilder();
```

```
            String line;
```

```
            while ((line = reader.readLine()) != null) {
```

```
                response.append(line);
```

```

    }

    reader.close();

    // Parse JSON data
    JSONObject jsonResponse = new JSONObject(response.toString());
    String weather =
jsonResponse.getJSONArray("weather").getJSONObject(0).getString("description");
    double temp = jsonResponse.getJSONObject("main").getDouble("temp");

    String result = "Weather: " + weather + "\nTemperature: " + temp + "°C";

    // Update the TextView on the main thread
    getActivity().runOnUiThread(() -> weatherInfo.setText(result));

    } catch (Exception e) {
        getActivity().runOnUiThread(() -> {
            Toast.makeText(getContext(), e.getMessage(), Toast.LENGTH_SHORT).show();
        });
    }
    }).start();
}
}

```

fragment_settings.xml:

```

<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="16dp"
    android:background="@drawable/background_image">

    <!-- Title for Settings -->

```

```

<TextView
    android:id="@+id/settings_title"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="@string/settings_title"
    android:textColor="@color/green"
    android:textSize="24sp"
    android:textStyle="bold"
    android:layout_marginBottom="16dp"/>

<!-- ListView to display settings options -->
<ListView
    android:id="@+id/settings_list"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:divider="@android:color/darker_gray"
    android:dividerHeight="0.5dp"
    android:padding="8dp"/>

</LinearLayout>

```

SettingsFragment.java:

```

package com.example.agroseva;

import android.os.Bundle;
import android.view.LayoutInflater;
import android.view.View;
import android.view.ViewGroup;
import android.widget.ListView;

import androidx.fragment.app.Fragment;

public class SettingsFragment extends Fragment {

```

```

private ListView settingsListView;

@Override
public View onCreateView(LayoutInflater inflater, ViewGroup container, Bundle savedInstanceState) {
    // Inflate the layout for the Settings fragment
    View view = inflater.inflate(R.layout.fragment_settings, container, false);

    settingsListView = view.findViewById(R.id.settings_list);

    // Define the options to display in the ListView
    String[] settingsOptions = {
        "FAQ",
        "Privacy Policy",
        "Terms and Conditions",
        "Return Policy",
        "Rate Us",
        "Logout" // Add Logout option here
    };

    // Create an adapter and set it to the ListView
    SettingsAdapter adapter = new SettingsAdapter(getContext(), settingsOptions);
    settingsListView.setAdapter(adapter);

    return view;
}
}

```

strings.xml:

```
<resources>
    <!-- App General -->
    <string name="app_name">Agro Seva</string>
    <string name="signup">SIGNUP</string>
    <string name="login">LOGIN</string>
    <string name="reset">RESET PASSWORD</string>
    <string name="email">Email</string>
    <string name="reset2">Reset</string>
    <string name="f_b">Forgot Password?</string>
    <string name="register">REGISTER</string>
    <string name="fullname">Full Name</string>
    <string name="username">Username</string>
    <string name="password">Password</string>
    <string name="cpassword">Confirm Password</string>
    <string name="old_account">Already have an account?</string>
    <string name="skip">Skip Login</string>

    <!-- Home Page -->
    <string name="homepage">Home Page</string>
    <string name="notification">Notification</string>
    <string name="back">Back</string>
    <string name="rate">Rate</string>
    <string name="store">Store</string>
    <string name="share">Share</string>
    <string name="aboutus">About Us</string>
    <string name="paymentinfo">Payment Info</string>
    <string name="settings">Settings</string>

    <!-- Reset Password -->
    <string name="resetcontent">Check your email for a link to reset your password securely.</string>

    <!-- Product Info -->
```

```

<string name="products">Products</string>
<string name="price">Price</string>
<string name="quantity">Quantity</string>
<string name="error_empty_quantity">Quantity is Empty</string>

<!-- Weather Section -->
<string name="weather">Weather</string>
<string name="translate">Translate</string>

<!-- Categories -->
<string name="fruits">Fruits</string>
<string name="vegetables">Vegetables</string>
<string name="fertilizer">Fertilizers</string>
<string name="machine_equipment">Equipment</string>
<string name="flowers">Flowers</string>
<string name="other_services">Other Services</string>

<!-- Sample Products -->
<string name="apple">Apple</string>
<string name="orange">Orange</string>
<string name="grapes">Grapes</string>
<string name="banana">Banana</string>
<string name="jackfruit">Jackfruit</string>

<!-- Onion Guide -->
<string name="oh">Onion Planting Guide</string>

<!-- Weather Section -->
<string name="wh1">Select City in Tamil Nadu</string>
<string name="wh2">Weather Information</string>
<string name="ws">Please select a city to get weather updates.</string>
<string name="get_weather">Get Weather</string>

```



```

<!-- Arrays -->
<string-array name="consultancy_types">
    <item>Soil Analysis</item>
    <item>Crop Selection</item>
    <item>Fertilizer Guidance</item>
    <item>Irrigation Planning</item>
    <item>Market Trends</item>
    <item>Pest Control</item>
</string-array>
<string-array name="crops_array">
    <item>Tomato</item>
    <item>Onion</item>
</string-array>
<string-array name="tamil_nadu">
    <item>Dindigul</item>
    <item>Madurai</item>
    <item>Theni</item>
    <item>Viruthunagar</item>
    <item>Karur</item>
    <item>Kanniyakumari</item>
</string-array>

<!-- Order Screen -->
<string name="order_summary">Order Summary</string>
<string name="total_amount">Total Amount: ₹0</string>
<string name="your_name">Your Name</string>
<string name="phone_number">Phone Number</string>
<string name="your_address">Your Address</string>
<string name="your_city_town">Your City/Town</string>
<string name="your_district">Your District</string>
<string name="pincode">Pincode</string>
<string name="proceed_to_payment">Proceed to Payment</string>

```

<!-- Vegetables Section -->

<string name="tomato">Tomato</string>

<!-- General Strings -->

<string name="date">Date</string>

<string name="task_name">Task Name</string>

<string name="number_of_workers">Number of Workers</string>

<string name="salary_per_worker">Salary Per Worker</string>

<string name="total_salary">Total Salary</string>

<string name="calculate_total_salary">Calculate Total Salary</string>

<!-- Hints -->

<string name="enter_number_of_workers">Enter Number of Workers</string>

<string name="enter_salary_per_worker">Enter Salary Per Worker</string>

<!-- General Strings -->

<string name="no_notification">No Notifications</string>

<!-- Common Strings -->

<string name="instruction">Instruction</string>

<string name="soil_test">Soil Test</string>

<string name="water_management">Water Management</string>

<string name="labor_tracking">Labor Tracking</string>

<string name="profit_loss">Profit/Loss Estimation</string>

!-- Section Titles -->

<string name="expenditures">Expenditures</string>

<string name="income">Income</string>

<!-- Expenditure Input Hints -->

<string name="hint_land_cost">Land Cost (₹)</string>

<string name="hint_depreciation">Depreciation (₹)</string>

<string name="hint_taxes">Taxes and Insurance (₹)</string>

<string name="hint_seeds">Seeds and Planting Material (₹)</string>

<string name="hint_fertilizers">Fertilizers and Soil Amendments (₹)</string>
<string name="hint_labor">Labor (₹)</string>
<string name="hint_water">Water/Irrigation (₹)</string>
<string name="hint_fuel">Fuel and Electricity (₹)</string>
<string name="hint_machinery_maintenance">Machinery Maintenance (₹)</string>
<string name="hint_transportation">Transportation and Logistics (₹)</string>
<string name="hint_storage">Storage and Packaging (₹)</string>

<!-- Income Input Hints -->

<string name="hint_revenue_crop_sales">Crop Sales (₹)</string>
<string name="hint_revenue_livestock">Livestock Products (₹)</string>
<string name="hint_revenue_by_products">By-products (₹)</string>
<string name="hint_revenue_value_added">Value-added Products (₹)</string>

<!-- Settings -->

<string name="settings_title">Settings</string>

<!-- Title -->

<string name="soil_test_title">Soil Suitability Test for Tomatoes</string>

<!-- Input Labels -->

<string name="enter_soil_moisture">Enter Soil Moisture</string>
<string name="enter_temperature">Enter Temperature</string>
<string name="enter_humidity">Enter Humidity</string>
<string name="enter_nitrogen">Enter Nitrogen (mg/kg)</string>
<string name="enter_phosphorus">Enter Phosphorus (mg/kg)</string>
<string name="enter_potassium">Enter Potassium (mg/kg)</string>
<string name="select_soil_type">Select Soil Type</string>

<!-- Button Text -->

<string name="check_suitability">Check Suitability</string>

<!-- Titles -->

<string name="title_text">Growing Healthy Tomatoes</string>

<string name="choose_variety_title">Choose the Right Variety</string>

<string name="planting_time_title">Plan the Planting Time</string>

<string name="location_title">Select the Location</string>

<string name="harden_off_title">Harden Off Seedlings</string>

<string name="pruning_title">Pruning</string>

<string name="harvesting_title">Harvesting</string>

<!-- Descriptions -->

<string name="choose_variety_details">Select tomato seeds based on your climate, soil type, and intended use (e.g., cherry tomatoes, heirloom, or Roma). Consider disease-resistant varieties for a healthier crop.</string>

<string name="planting_time_details">Tomatoes thrive in warm weather. Plan to plant after the last frost date in your region. Use a calendar to ensure your seedlings are ready for transplanting.</string>

<string name="location_details">Choose a location with full sun (at least 6-8 hours of direct sunlight daily). Ensure good air circulation to prevent fungal diseases.</string>

<string name="harden_off_details">1. Dig a hole twice as wide and slightly deeper than the seedling's root ball.\n\n2. Plant deeply - bury the stem up to the first set of leaves to encourage strong root development.\n\n3. Space plants 18-24 inches apart.\n\n4. Water thoroughly after planting.</string>

<string name="pruning_details">1. Remove small shoots growing between the main stem and branches to focus energy on fruit production.\n\n2. Prune lower leaves to prevent disease spread from the soil.</string>

<string name="harvesting_details">1. Tomatoes are ready to harvest when they are firm, fully colored, and slightly soft to the touch.\n\n2. Gently twist or cut the fruit from the plant to avoid damage.\n\n3. Harvest regularly to encourage continued fruit production.</string>

<!-- Section Titles -->

<string name="oh1">Seed Selection</string>

<string name="oh2">Soil Preparation</string>

<string name="oh3">Planting</string>

<string name="oh4">Watering</string>

<string name="oh5">Harvesting</string>

<!-- Section Details -->

<string name="opara1">Choose high-quality onion seeds that are suited to your climate and soil

conditions. Opt for certified seeds for better results.</string>

<string name="opara2">Prepare the soil by loosening it to improve aeration. Add organic compost and ensure the soil has good drainage to prevent waterlogging.</string>

<string name="opara3">Plant onion seeds or seedlings in rows, leaving enough space between them to allow for growth. Ensure they are planted at the appropriate depth.</string>

<string name="opara4">Water the onions regularly, especially during dry periods. Avoid overwatering, as onions prefer moist but not soggy soil.</string>

<string name="opara5">Harvest onions when the tops turn yellow and start to fall over. Cure the onions in a dry, well-ventilated area for better storage.</string>

<!-- Section Titles -->

<string name="onion">Onion</string>

<!-- Common Text Strings -->

<string name="select_category">Select Category:</string>

<string name="select_crop">Select Crop:</string>

<string name="select_date">Select Date:</string>

<string name="select_time">Select Time:</string>

<string name="calculate_profit_loss">Calculate Profit / Loss</string>

<!-- Weather Related Strings -->

<string name="weather_select_city">Select City:</string>

<string name="weather_info_label">Weather Info:</string>

<string name="weather_default_info">No weather data available</string>

<string name="get_weather_button">Get Weather</string>

<string name="login2">Login</string>

<string name="register2">Register</string>

<string name="register3">Register Account</string>

<string name="th">Tomato Planing Guide</string>

<array name="task_names">

<item>Task 1</item>

<item>Task 2</item>

```
<item>Task 3</item>
</array>
```

```
<array name="soil_types">
  <item>Clay</item>
  <item>Sandy</item>
  <item>Loam</item>
</array>
```

```
<array name="category_array">
  <item>Vegetables</item>
  <item>Fruits</item>
  <item>Crops</item>
</array>
```

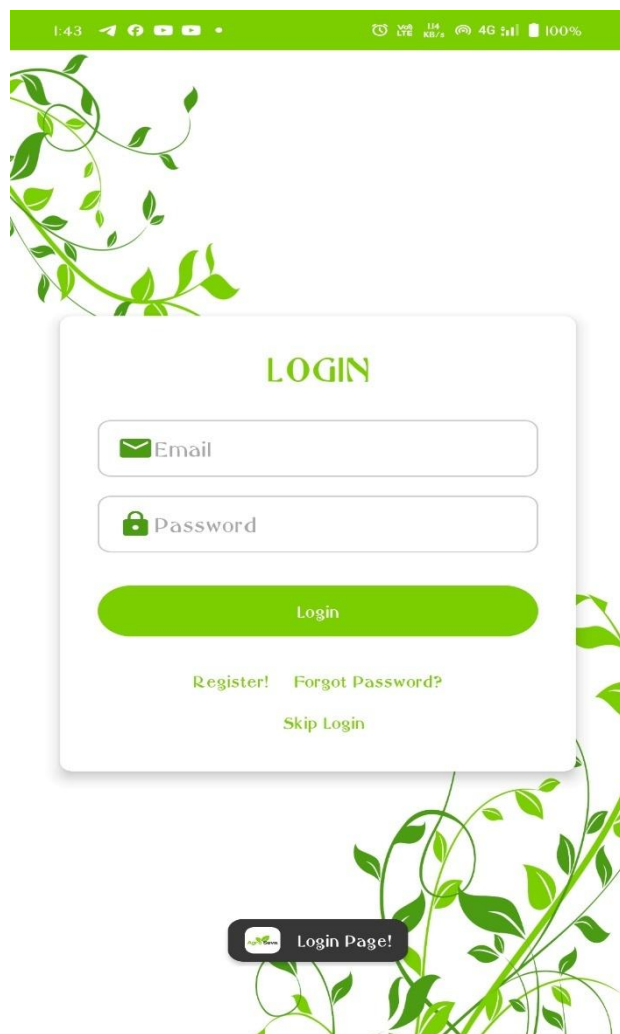
```
</resources>
```

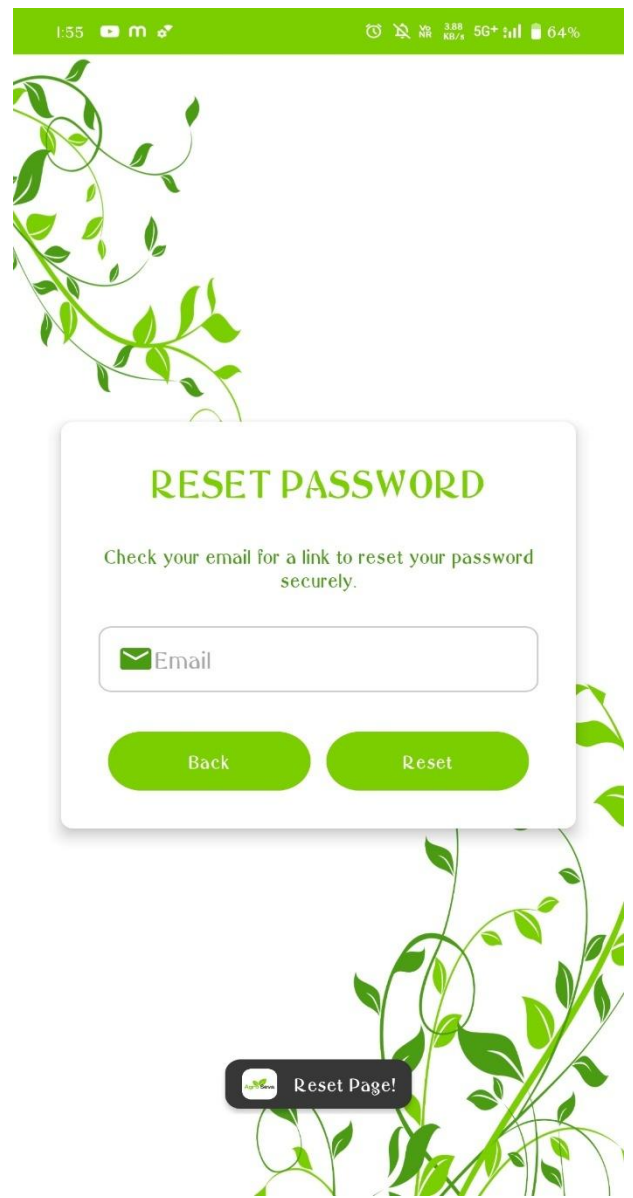
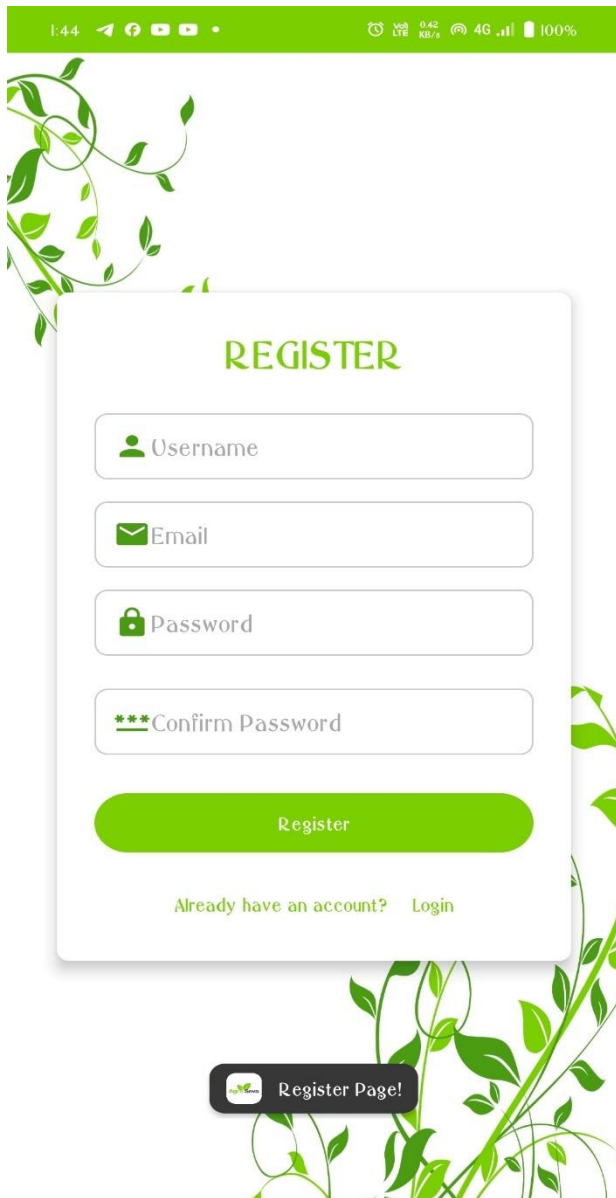
colors.xml:

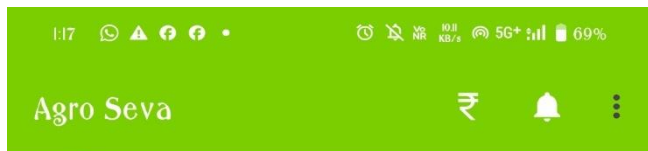
```
<?xml version="1.0" encoding="utf-8"?>
<resources>
  <color name="black">#FF000000</color>
  <color name="color_white">#FFFFFF</color>
  <color name="color_primary">#4CAF50</color>
  <color name="green">#8fcc2d</color>
  <color name="green2">#5f992f</color>
  <color name="primary">#FF6200EE</color>
  <color name="appBackground">#8fcc2d</color>
</resources>
```

APPENDIX B – SAMPLE SCREENSHOT

APP SCREENSHOT:







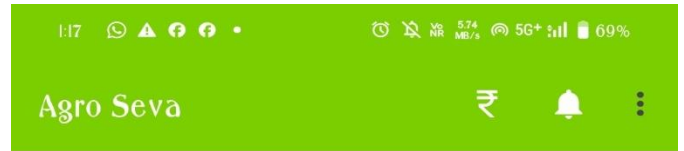
Products



Vegetables



Fruits

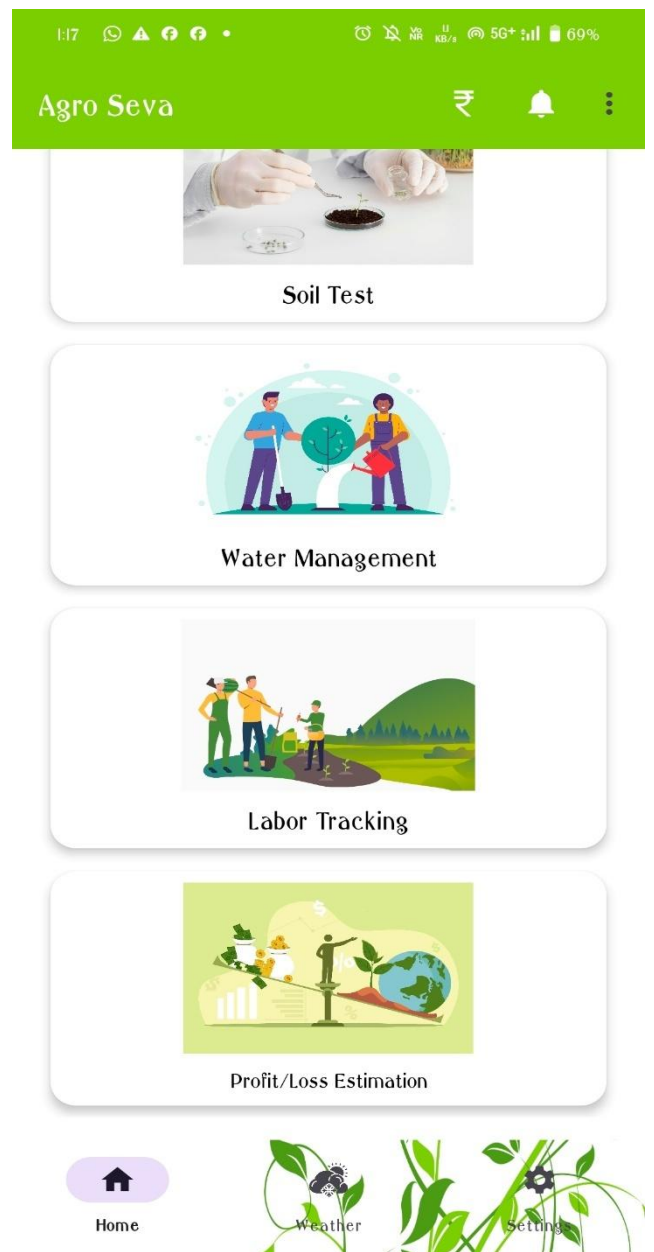
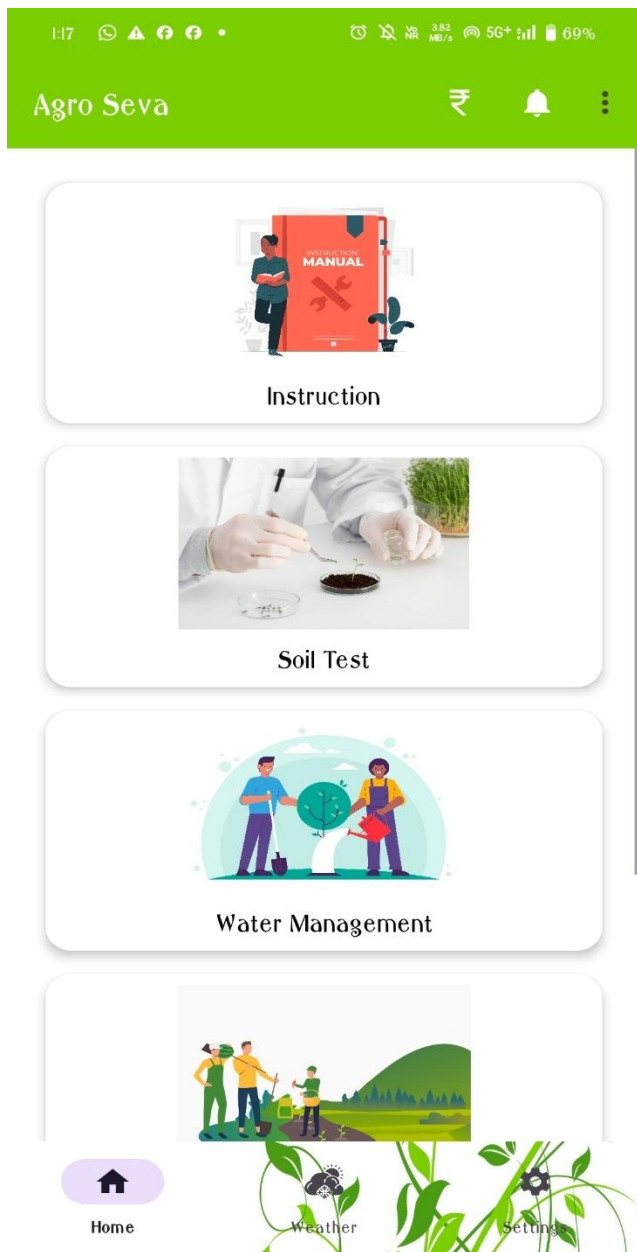


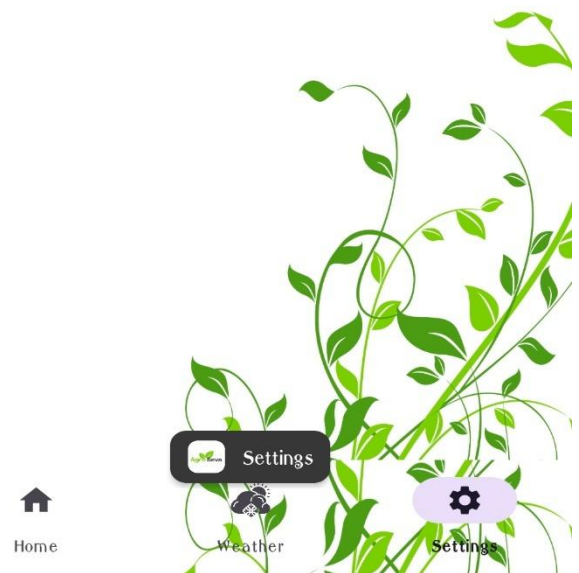
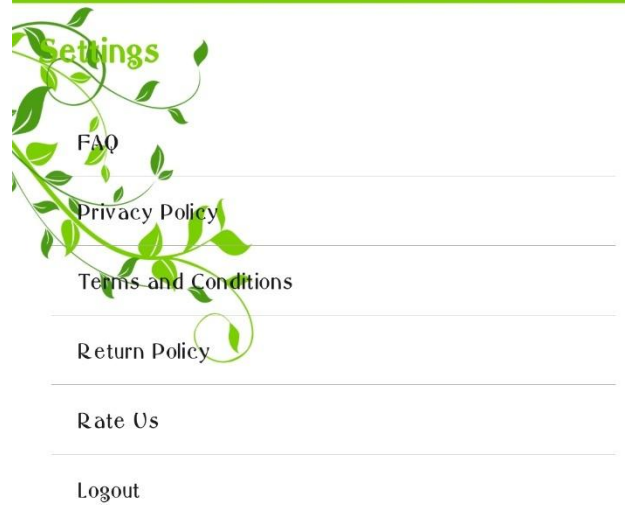
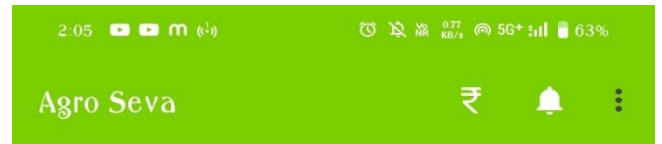
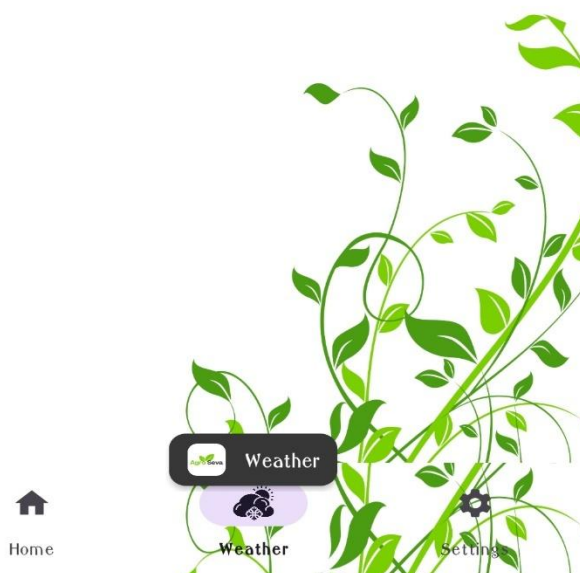
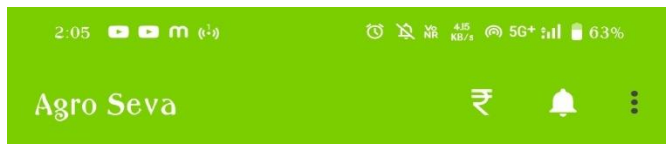
Vegetables



Tomato







CHAPTER VIII

CONCLUSION

CONCLUSION

Farm Management Agro-Seva represents a step forward in integrating technology into agriculture, specifically tailored for the needs of farmers cultivating tomatoes. The app simplifies complex agricultural tasks by offering a variety of modules, including soil testing, step-by-step cultivation instructions, water reminders, and financial analysis for profit and loss. These features ensure that farmers can make data-driven decisions to improve productivity and maximize their yield.

The inclusion of labour tracking provides an efficient way to manage workforce details, while the Offer Notification feature keeps users informed about valuable discounts and government schemes. The language translator enhances accessibility, making the app usable by a diverse audience in their preferred language, such as Tamil or English.

The app integrates advanced tools like a weather API to provide real-time updates, enabling farmers to adapt their plans based on climatic conditions. The rate module tracks daily market prices, ensuring farmers can make informed decisions about selling their produce for maximum profit. Additionally, notification alerts keep users updated on critical information, including weather changes, market trends, and app updates.

By categorizing the app into vegetables and fruits, with a detailed focus on tomatoes, it ensures simplicity in navigation and usability. Farmers gain access to comprehensive information and guidance, helping them address key challenges in modern agriculture.

Farm Management Agro-Seva combines traditional farming knowledge with cutting-edge technology, bridging the gap to create a sustainable and productive future for agriculture. This app is a practical and user-friendly tool that empowers farmers to enhance their practices, reduce costs, and increase profitability while staying informed about weather and market trends, contributing significantly to the agricultural sector's growth.

CHAPTER IX

FUTURE ENHANCEMENT

FUTURE ENHANCEMENT

Farm Management Agro-Seva is set to evolve further with planned enhancements that bring advanced technology to the forefront of agriculture. Future updates aim to include AI-powered crop recommendations based on soil, weather, and market conditions, along with drone integration for efficient monitoring and pest control. Blockchain technology will be leveraged for transparent supply chain management, while AI-based pest and disease identification tools will offer quick remedies. Expanding the app's scope, new crop modules, a community forum for farmers, and financial tools like loan calculators and subsidy trackers will be introduced. Offline functionality will ensure usability in remote areas, and an integrated e-marketplace will connect farmers directly with buyers, maximizing profits. These innovations, coupled with sustainability analytics, aim to empower farmers, enhance productivity, and foster a sustainable agricultural future.

CHAPTER X

BIBLIOGRAPHY

BIBLIOGRAPHY:

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